

the core clock by 50-100 MHz based on what card you have. Run a benchmark to test for stability and then go back to increasing the core clock. Rinse and repeat till there are stability issues or if you start to see artifacts.

Since there is a lot of visual feedback while overclocking graphics cards you'll instantly come to know if anything's wrong with your current settings. That's when you bump up the memory clock because the artifacts indicate insufficient bandwidth. It could also indicate that your core clock is a bit too high. As you go through a permutation of settings, you will figure out which one is the culprit. Ideally, you don't want memory bandwidth to be a bottleneck, which is why you'll want to bump the memory

clock to the most that's possible. If you don't want to go through the entire process of hunting for the right combination of core and memory clock, then you can always rely on the OC scanner feature and let it do the overclocking for you. The entire process takes about 20 minutes and you'll get a fairly decent overclock at the end.

CUSTOM FIRMWARE FLASHING

While not technically overclocking, you can flash a custom firmware onto your graphics card and do away with the software part altogether. Sometimes, different graphics cards are different only in terms of their firmware while the underlying hardware remains the

same. You can take the firmware of a Radeon RX 570 and flash it onto a Radeon RX 470. This does not apply to all RX 470 cards. NVIDIA Founders Edition (FE) cards have power limits baked into them, flashing the BIOS of an aftermarket card can help you remove those power limits. This was the case with the GTX 1080 Ti Founders Edition. You could flash the ASUS STRIX 1080 Ti BIOS onto the 1080 Ti FE and enable higher overclocks. The flashing process uses a simple CLI and uses nvflash. Radeon cards use ATIFlash for the same purpose. You can always find graphics card BIOS from aftermarket cards on websites such as TechPowerUp. Do note, flashing a BIOS onto your card will invalidate your warranty. 

PERIPHERAL OVERCLOCKING GUIDE

Ramping up polling rates and refresh rates

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The word overclocking is primarily associated with bumping up the clock speeds of components including the CPU, GPU and RAM. However, when you use 'overclocking' with other components or devices, the meaning changes. In our pursuit of achieving better performance, we are now exploring the potential of our peripherals that include mice, keyboards and monitors. Overclocking these peripherals

aren't about clock speeds but a different parameter. So, overclocking in a mouse or keyboard is increasing the polling rate whereas, in a monitor, you are increasing the refresh rate of the display. This guide will help you overclock your peripherals but do note, it could render your device unusable, so proceed with caution.

OVERCLOCKING YOUR MOUSE

Things to know

Before tweaking up the polling rate of your mouse, it is a good idea to

understand what the polling rate is. While your mouse is connected to your PC, they both communicate with each other at a particular interval of time. This interval is limited by the hardware of the mouse. The polling rate of the mouse is measured in Hz. You will generally find mice with a polling rate of 125Hz or 8 milliseconds. So, the data from the mouse is updated after every 8 milliseconds. More expensive mice or rather, gaming mice ship with a higher polling rate of either 500Hz or 1,000Hz. Nowadays, most of the gaming mice out there come with a default polling rate of 1,000Hz or 1ms. This means that there's only going to be a